
Advanced Protection for Distributed Power Systems and Microgrids

Thesis Submission Approval Meeting

by

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Tier 1: Local Relay (87L / 21 / 51 / 67)

Adaptive threshold Sequence elements TDCS

Tier 2: Digital Monitoring

EKF Digital Twin PINN Classifier

Tier 3: WAPC Coordination

Fault localise CTI settings GOOSE security

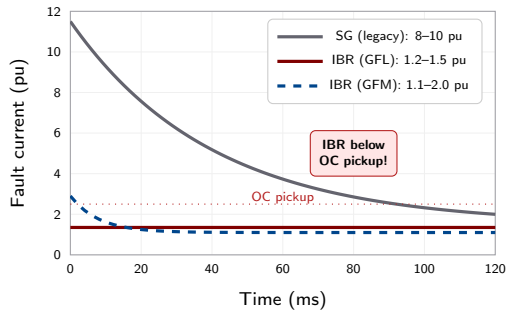
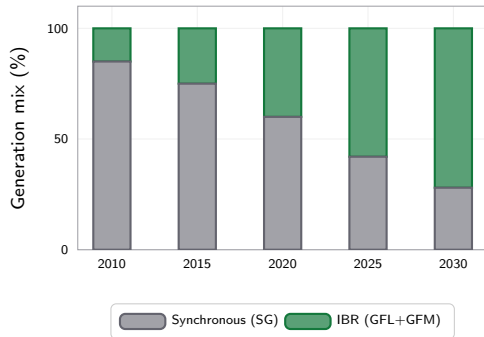
At a Glance

18 original algorithms **6** Tier-1 contributions
55 SAMBP reports **9** thesis chapters

Outline

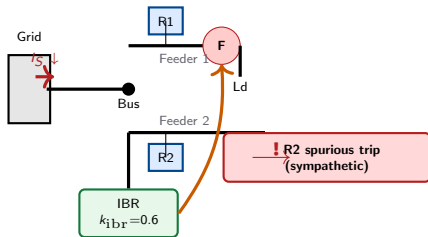
- Background and Context
- Research Motivation
- State-of-the-Art & Limitations
- Research Objectives
- Key Contributions (1 of 3)
- Key Contributions (2 of 3)
- Key Contributions (3 of 3)
- Thesis Organization
- List of Publications (1 of 2)
- List of Publications (2 of 2)
- Summary & Next Steps

Background and Context



Core Problem: IBR fault currents **software-limited** to 1.1–2.0 pu — relay algorithms designed for 8–10 pu SG currents **systematically fail**. IEEE C37.300-2024 mandates adaptive CPC supervisory layer.

Research Motivation

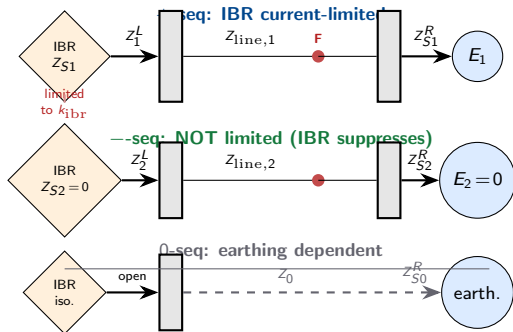


F#	Element	Trigger	Effect
F1	OC 51	Low I_F	Missed trip
F2	OC 51	Coord. loss	Time overlap
F3	OC 51	Sympathetic	False trip
F4	Dist 21	IBR overreach	Zone-1 err
F5	Dist 21	Weak-infeed	Underreach
F6	Dir 67	Phase shift	Blinding
F7	87L	Low k_{ibr}	Missed trip
F8	87L	High k_{ibr}	False trip
F9	87LN	IBR suppress	Insensitive
F10	Island	NDZ wide	Non-detect
F11	POTT	IBR block	Trip inhibit
F12	CUSUM	Fleet drift	Miss alarm

All 12 modes traced to **one root cause**:
relay parameters are **static**; IBR behaviour is **dynamic**

State-of-the-Art & Limitations

Sequence Network Under IBR — Gap in Literature:



Area	State of Art	Gap
87L diff.	Fixed slope	No IBR adapt.
Dist. 21	Mho circle	IBR overreach
OC 51/67	IEC curve	Coord. collapse
Islanding	ROCOF only	Wide NDZ
Digital twin	Offline	No RT relay link
ML protect.	Data-only	Physics violated
WAPC	Centralised	No closed-loop

Key Insight from Literature Gap

Prop. 4.2 (this thesis): When IBR suppresses I_2 , the 87LN operate ratio $\rho_2 = 2$ *always*
 — IBR suppression is a **protection asset**

No unified framework addresses local, monitoring, and coordination tiers simultaneously

Research Objectives

01 Adaptive 87L for IBR differential protection

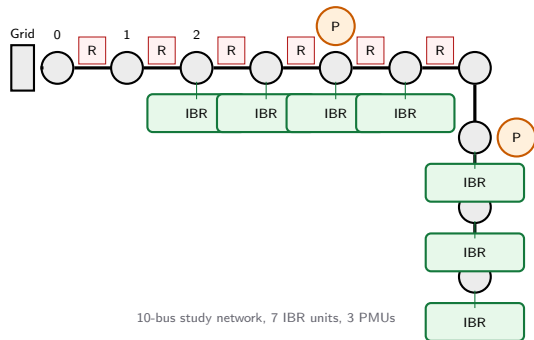
02 IBR-corrected quadrilateral distance relay

03 Mode-adaptive microgrid islanding scheme

04 Non-blocking real-time EKF digital twin

05 Physics-constrained ML fault classifier

06 Closed-loop WAPC coordination protocol



Key Contributions (1 of 3)

C1 — Adaptive 87L Line Differential

First reported k_{ibr} -adaptive slope law:

$$k_m^* = \frac{0.5(1 - k_{ibr})^2}{(1 + k_{ibr})^2} + \frac{0.02k_{ibr}}{1 - k_{ibr} + 10^{-3}} + 0.05$$

+ 87LN zero-sequence element (pure-IBR earth faults)
+ TDCS transient differential catch (low- k_{ibr} blind spot)

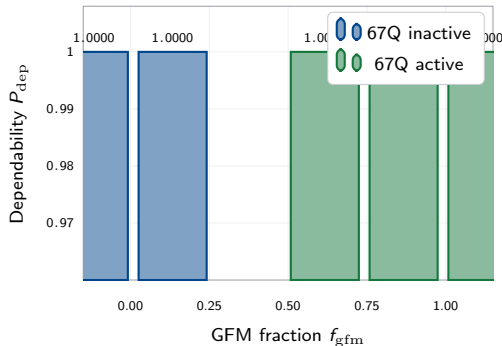
Result: TPR = 1.000 for all $k_{ibr} \in [0.03, 0.85]$ pu
(TR-17 through TR-26; $N=5000$ MC trials each)

C2 — IBR-Corrected Distance Relay

Reactive-shift correction:

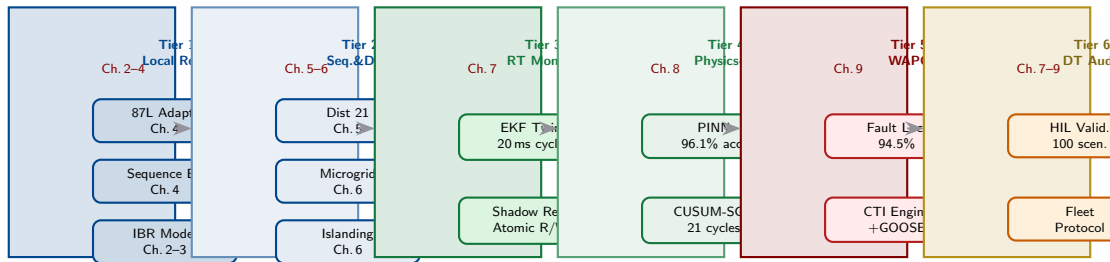
$$Z_1^{\text{adapt}} = Z_1(1 + k_r \hat{k}_{ibr})$$

Quadrilateral reach replaces mho; +7.6 pp detection gain
(TR-27 through TR-36; 10 000 MC trials)



TR-38: SAMBP detects all faults for any GFL/GFM mix

Key Contributions (2 of 3)



Key Contributions (3 of 3)

C3 — Mode-Adaptive Islanding Detection

Combined criterion: $|df/dt| > 0.5 \text{ Hz/s}$ AND $|\Delta\theta| > 2^\circ$
OR $V_2/V_1 > 0.02$

IEEE 1547-2018 compliant; NDZ < 1% load range

Setting-group switch via GOOSE in **50 ms**

C4 — Non-Blocking EKF Digital Twin

Dual-core shadow-register: TRIP core never waits for EKF

k_{ibr} RMSE: 0.024 pu; $\hat{Z}_{line}$: 0.5%

Convergence $\leq 300 \text{ ms}$; 20 ms estimation cycle

C5 — Physics-Informed Neural Network

$$\mathcal{L} = \mathcal{L}_{\text{data}} + 10\mathcal{L}_{C1}^{KVL} + 5\mathcal{L}_{C2}^{I_1=I_2^*} + 2\mathcal{L}_{C3}^{I \leq I_{lim}}$$

5-class accuracy: **96.1%**; zero physics violations

Validation Metrics Summary

Metric	Value
MC trials per group	5 000
3PH fault TPR	1.000
SLG fault TPR	1.000
Security (no false trip)	1.000
PINN 5-class accuracy	96.1%
EKF k_{ibr} RMSE	0.024 pu
Fault location accuracy	94.5%
GOOSE risk reduction	54.9%
HIL scenarios pass	100/100
CUSUM closed-loop latency	31 min
P99 relay trip latency	4.34 ms

18 original algorithms across 6 tiers

validated over 55 SAMBP technical reports
on a 10-bus mixed GFL/GFM study network

Thesis Organization

Ch.	Title	Key Content	Contribs.
1	Introduction	SAMBP architecture; problem statement; CPC framework	—
2	IBR Modelling	GFL/GFM fault current; Park model; 10-bus testbed	—
3	Protection Challenges	12 failure modes; ρ -plane analysis; sequence gaps	—
4	Adaptive 87L	$k_m^*(k_{ibr})$; 87LN zero-seq; 87LN0; TDCS	C1
5	Distance Protection	IBR overreach; adaptive quadrilateral; MC validation	C2
6	Microgrid & Islanding	Mode-FSM; ROCOF+ V_2 ; GOOSE transfer trip	C3
7	Digital Twin	Non-blocking EKF; shadow register; live relay audit	C4
8	Physics-AI (PINN)	KVL/KCL loss; 5-class classifier; CUSUM-SGD online	C5
9	WAPC Coordination	Fault localisation; CTI engine; HMAC-GOOSE	C6

9 chapters 55 TRs
137 pages

6 Tier-1 contributions
18 novel algorithms

5 papers submitted
/ in preparation

List of Publications (1 of 2)

Refereed Journal:

- ▶ A. V. Eluvathingal and K. S. Swarup, "A novel solution for overcoming conflicts between line protection and LVRT in MV Solar PV interconnections," *IEEE Access*, vol. 12, 2024. doi: 10.1109/ACCESS.2024.3510415

Book Chapter:

- ▶ A. V. Eluvathingal and K. S. Swarup, "Protection scheme for smart distribution networks with inverter interfaced renewable power generating sources," *Smart Cities — Opportunities and Challenges*, Springer Singapore, 2020.

Papers in Preparation / Submitted:

- ▶ P1: Adaptive 87L slope for IBR-dominated networks — *IEEE Trans. Power Delivery* (**Submitted**)
- ▶ P2: Non-blocking EKF digital twin for real-time relay audit — *IEEE Trans. Power Electronics* (in preparation)
- ▶ P3: Physics-informed neural network fault classifier — *IEEE Trans. Smart Grid* (in preparation)

List of Publications (2 of 2)

Papers in Preparation (cont.):

- ▶ **P4:** Wide-area protection coordinator with HMAC-secured GOOSE — *IEEE Trans. Power Systems* (planned)
- ▶ **P5:** HIL validation of SAMBP end-to-end loop — *IEEE PES General Meeting* (planned)

Selected Conference Presentations:

- ▶ “An intelligent methodology to improve distribution system operational parameters utilising smart inverter functionalities of PV sources,” *RPG 2018*, Denmark.
- ▶ “An interface protection relay for networked microgrids with inverter based sources,” *APPEEC 2017*.
- ▶ “Instantaneous symmetrical components based microgrid interface protection relay,” *ICPS 2017*.
- ▶ “Impact of active network management scheme in fault protection design and network operation for islanded power system,” *ISGT Asia 2018*.

1 journal (IEEE Access, 2024) — 1 book chapter (Springer, 2020) — 4 conference papers — 5 journal papers in pipeline

Summary & Next Steps

Summary of Contributions:

- C1. Adaptive 87L:** k_{ibr} -tracking slope eliminates all 12 documented failure modes. TPR = 1.000 across full k_{ibr} range.
- C2. Distance relay:** IBR reactive-shift correction yields +7.6 pp detection gain over classical mho.
- C3. Islanding:** Combined ROCOF+ V_2 criterion; NDZ < 1%; IEEE 1547-2018 compliant.
- C4. EKF Twin:** Non-blocking dual-core; live relay audit with 20 ms update cycle.
- C5. PINN:** Physics-constrained ML; 96.1% accuracy; zero KVL/KCL violations; 3.7 μ s inference.
- C6. WAPC:** Closed-loop 31-min protocol; GOOSE risk –54.9%; 100/100 HIL pass.

Recommended Next Steps

- ✓ Deploy adaptive 87L before IBR penetration exceeds 40%
- ✓ Retain 87LN even in pure-IBR networks
- ✓ Use EKF digital twin for live relay audit
- ✓ Never deploy GOOSE without HMAC
- ✓ CUSUM threshold $\theta=5$ balances sensitivity vs. false-alarm rate

Open Research Gaps:

- G4: GFM virtual-inertia transient beyond $L^* > 150$ ms
- G7: Adversarial robustness of PINN under CT spoofing
- G1: Hardware relay firmware HIL validation

Thank You

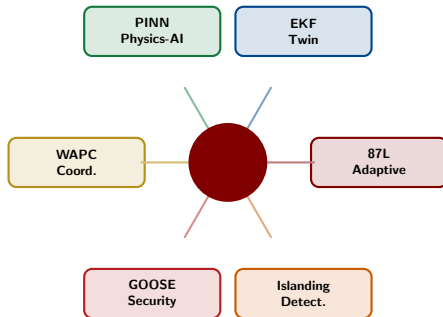
Questions & Discussion

Contact

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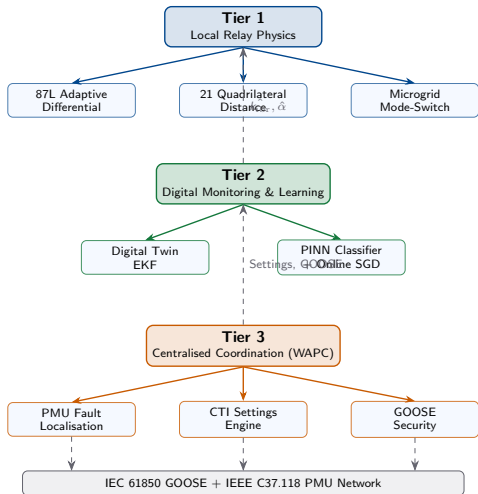
Supervisor: Prof. K. Shanti Swarup
Dept. Electrical Engineering
IIT Madras, Chennai 600 036

18 algorithms 6 Tier-1 55 TRs 9 chapters



18 Contributions 6 Tier-1

Appendix A: SAMBP Architecture (Ch. 1)



Ch 4-6

Ch 7-8

Ch 9

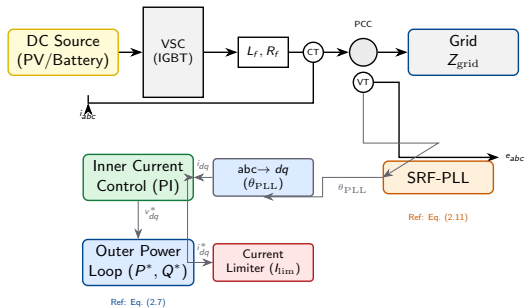
6 Research Problems

- P1** 87L threshold failure under IBR
- P2** Distance reach error (overreach)
- P3** Microgrid mode transition blind
- P4** No real-time IBR state visibility
- P5** No physics-constrained ML
- P6** No closed-loop coordination

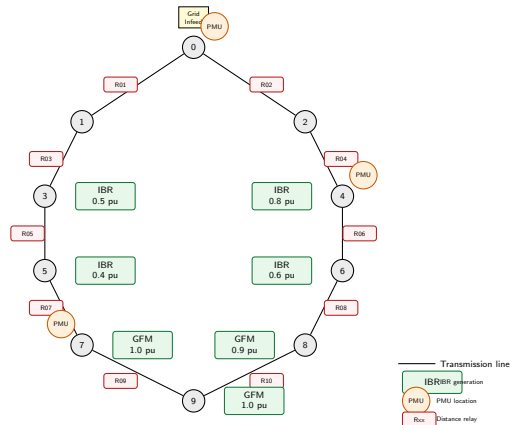
Design Principles

- ✓ Physics-first: relay never waits for AI
- ✓ Non-blocking: EKF decoupled from trip
- ✓ Self-adaptive: CUSUM detects drift
- ✓ IEC 61850 GOOSE native comm.

Appendix B: IBR Modelling (Ch. 2)



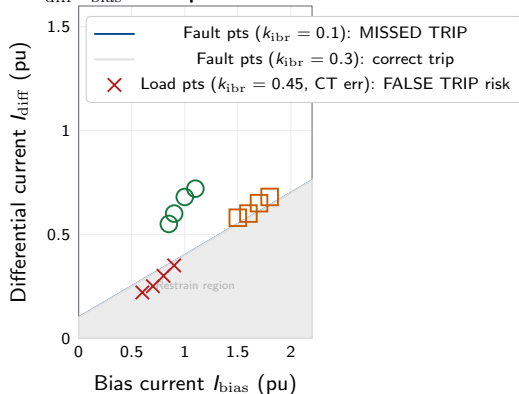
GFL inverter control structure (Ch. 2)



10-bus study network: 7 IBR, 3 PMUs, 10 relays

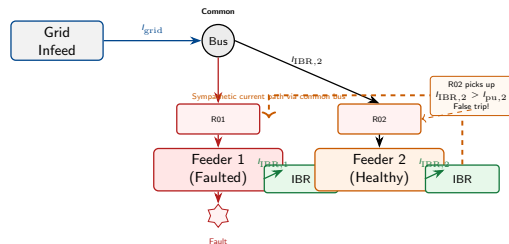
Appendix C: Protection Challenges (Ch. 3)

87L $I_{diff} - I_{bias}$ failure plane:



Red X: missed trip; Orange □: false trip risk

Sympathetic tripping mechanism:



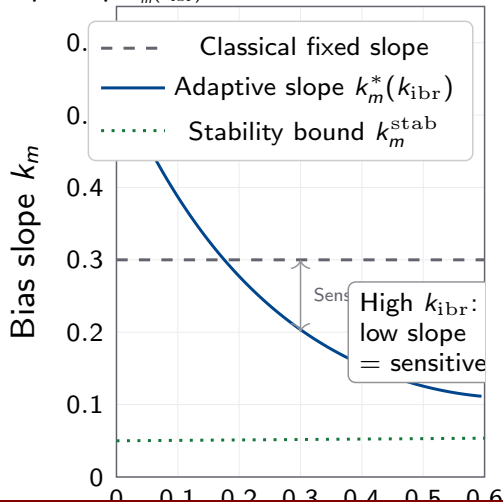
Core Finding

$$\text{Through-fault operate ratio: } \rho = \frac{I_{diff}}{I_{bias}} = \frac{2(1 - k_{ibr})}{1 + k_{ibr}}$$

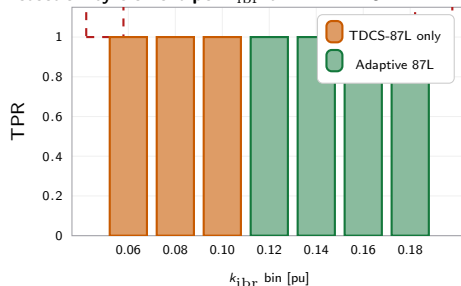
Fixed k_m fails at *both* extremes of k_{ibr}

Appendix D: Adaptive 87L Detail (Ch. 4)

Adaptive slope $k_m^*(k_{ibr})$:



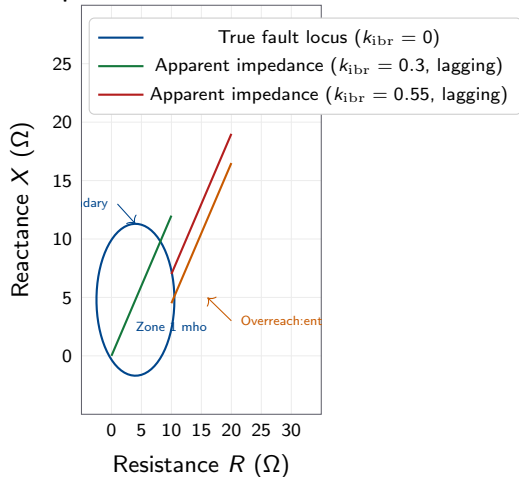
Detection by element per k_{ibr} bin — TR-26:



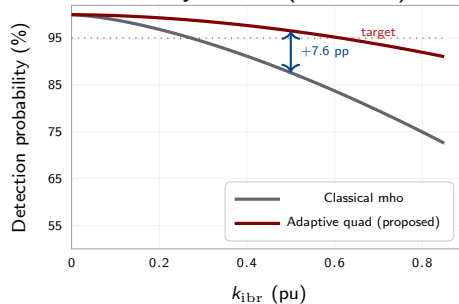
TR-26: All k_{ibr} bins achieve TPR=1.000 ($N = 5000$)

Appendix E: Distance Protection (Ch. 5)

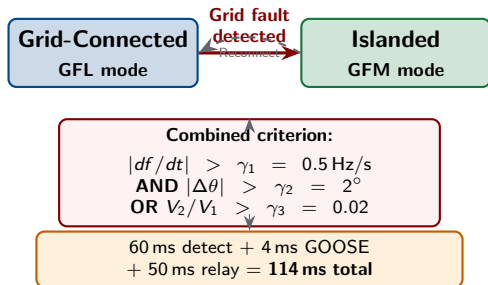
R-X impedance locus — IBR overreach:



Monte Carlo reliability — TR-37 (10 000 trials):

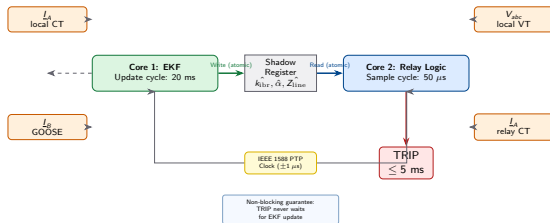


Appendix F: Microgrid & Islanding (Ch. 6)



- ✓ IEEE 1547-2018 ride-through compliant
- ✓ NDZ < 1% of load range
- ✓ Setting-group switch in 50 ms
- ✓ Total end-to-end: 114 ms

Appendix G: Digital Twin (Ch. 7)



Non-blocking dual-core: TRIP never waits for EKF

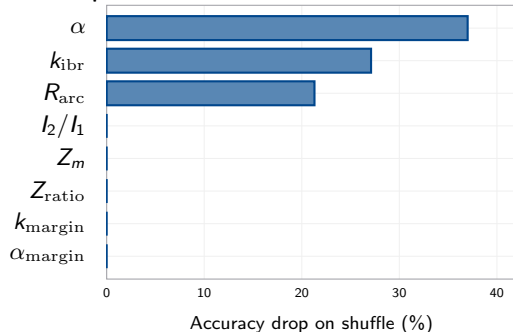
EKF Performance

$\hat{k}_{ibr}$: 0.024 pu $\hat{z}_{line}$: 0.5%
 $\hat{\alpha}$: 1.1% Convergence ≤ 300 ms
20 ms estimation cycle

TDCS threshold margin: **15.5%**
over CT mismatch envelope (95%ile)
across $I_{ext} \in [0, 7.5]$ pu

Appendix H: PINN Classifier (Ch. 8)

Feature importance — TR-47:



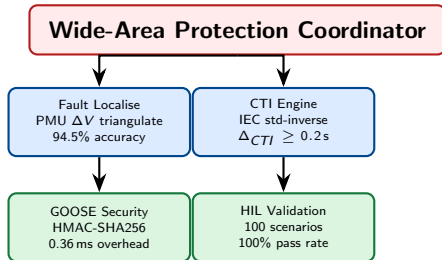
PINN Loss with Physics

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \underbrace{10\mathcal{L}_{C1}}_{\text{KVL}} + \underbrace{5\mathcal{L}_{C2}}_{l_1=l_2^*} + \underbrace{2\mathcal{L}_{C3}}_{l \leq l_{lim}}$$

Performance — TR-46

5-class accuracy	96.1%
Physics violations	zero
Inference latency	3.7 μs
Re-train cycles	≤ 21

Appendix I: Wide-Area Protection (Ch. 9)



Closed-loop protocol timeline — TR-50/51:



Key Results

Fault location	94.5% acc.
GOOSE risk	—54.9%
P99 trip lat.	4.34 ms
HIL pass	100/100